A SCHEME LOCALLY OF FINITE TYPE OVER A FIELD IS JACOBSON WITH ALGEBRAIC RESIDUE FIELDS AT CLOSED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author gives two independent routes to the result, the second by reference to the morphisms chapter; both sentences are reproduced verbatim and the cited results are carried as navigation stubs.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-locally-finite-type-Jacobson is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Jacobson property is local on an affine open covering. Properties, Lemma properties-lemma-locally-jacobson and Definition properties-definition-jacobson, invoked at proof lines 2682-2684 and named in the statement. Supplies the reduction to the affine case for both properties.
- Hilbert Nullstellensatz. Algebra, Theorem algebra-theorem-nullstellensatz, invoked at proof lines 2686-2688. Supplies the affine case of both assertions.
- Jacobson schemes and finite type points in the morphisms chapter. Morphisms, Lemmas morphisms-lemma-jacobson-finite-type-points, morphisms-lemma-Jacobson-universally-Jacobson and morphisms-lemma-ubiquity-Jacobson-schemes, invoked at proof lines 2689-2692. Supplies the author's second, independent route to the same conclusion.

Source reference. Original theorem 00FV: sources/stacks-chow/tags.txt, source lines 585–585. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02J5: sources/stacks-chow/tags.txt, source lines 3324–3324. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TB: sources/stacks-chow/tags.txt, source lines 2368–2368. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 02J6: sources/stacks-chow/tags.txt, source lines 3325–3325. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01P2: sources/stacks-chow/tags.txt, source lines 2215–2215. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01P4: sources/stacks-chow/tags.txt, source lines 2217–2217. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a scheme which is locally of finite type over k. Then*

(1) *for any closed point $x \in X$ the extension $\kappa(x)/k$ is algebraic, and*
(2) *X is a Jacobson scheme (Properties, Definition 01P2).*

Proof. A scheme is Jacobson if and only if it has an affine open covering by Jacobson schemes, see Properties, Lemma 01P4. The property on residue fields at closed points is also local on X. Hence we may assume that X is affine. In this case the result is a consequence of the Hilbert Nullstellensatz, see Algebra, Theorem 00FV. It also follows from a combination of Morphisms, Lemmas 01TB, 02J5, and 02J6. □